Lecture 3

Wednesday, 18 January 2023 8:30 AM

Don't forget about books → about specific topics

Literature Sources

+ Primary → published first time & peer reviewed
+ Secondary → reddit, wiki → not cited in papers
+ Gray
  - ArXiv, conference proceedings, Thesis & dissertations
  - Research reports, fact sheets,
  - Review articles

On going process → at least 1 hr daily
→ Make email digest to get daily updates on research

Search Types

+ Keywords (author generated
+ Boolean operators (AND, NOT, OR)
+ Author (specific syntax)
+ Cited reference source → references mentioned in a good article
  → articles that use that specific references

Evaluating Sources

+ University reputation
+ Journal impact factor (Top 10 journals in each research field)
+ # cited of reference (filter by citations)

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+ Conflict of interest (company publishing data)

How to start?

+ Use multiple search engines!
  - Google scholar (Alerts using gmail account)
  - Web of science
  - SciFinder Scholar → natural language search (english sentence search)

Refining the search

+ Date
+ Review articles → and then interested articles

ORCID → unique identifier for each author → ResearcherID

Purdue library → online
→ good for books
→ catalogous & create account

HW: + Coming w/ keywords
+ Top 10 hits by 3 search tools } Screenshot & compare the results.

Lecture 4

Monday, 23 January 2023 8:32 AM

Qualifying Exam:

+ Written Report : Summary — Due: Sept. 15
  Intro
  Research Objective
  Literature Review & Theory
  Research plan
  Materials & Methods
  Preliminary Results
  Plan for next 12 months

+ Oral Report : 20 min (Uninterrupted) — Due: Oct. break!
  No back-up slide
  30 min of questions → general engineering knowledge
  → Details about methods

+ Research hypothesis (Problem definition)
  → Do not be vague
  → Define the rationale (why do you think that (your expectation) happens?)
  → Should be testable
  → justification of why you think you could do / falsify

Become skeptical → Why? How?

Hypothesis vs. problem definition

+ Evaluation : Current state-of-the art in field
  1) Defend underlying basis (why are you doing what you are doing)

  2) Impact of the proposed work : Why is this imp. to other ChemE's?
     Why should people care? → Read review papers to gain better understanding
     How does your work fit into

  3) Preliminary Results & Mastery of key Research Techniques :
     → point out the limitations of your approach/idea → and what you are going to do to deal w/ it
     → design of experiment
     → Independence
     → why & how (the machine works & what the results mean?)
     → does the data prove or disprove the initially-proposed hypothesis

4) Potential of the project for Tangible Product generation
- → Trajectory (Research plan)
- → Can change direction / report failure
- → Careful analysis of why things went wrong

5) Communication Ability:
- → Explain to other chemE's
- → Graph → clear, easily readable
- → Animations
- → enthusiastic voice

3 faculty; Grading: 2 "not passing" on 2 same fields → fail

Written: 95% pass rate

Oral: 75% " for first time

Full-page feedback, retake w/ 3 new faculty

Lecture 5

Wednesday, 25 January 2023 8:32 AM

Evaluation of Hypothesis

Process: 1) Hypothetical thinking → variables

2) Prediction

3) Testibility / Observables

4) Simple & clear hypothesis → should stand by itself.
↓
But also specific

5) Consistency

Do not prove your hypothesis
↓
Disproving is easier/better and then provide alternate hypothesis
↓
Practice going through w/ people [

Hypothesis study:

1) Specific → figures of merit, "create particles across several length

2) Testibility → ?

Fix: better
Conductive

Lecture 6

Monday, 30 January 2023 8:31 AM

Reading Articles

1) Peer reviewed sources → look at impact factor of journals (look at highly cited journals)

2) Order of authors: first → wrote & most contribution

Corresponding authors → often permanent address (or PI)

Others → small contributions (significant enough to get some data), meaningfull

3) Highlights / Content sections.

4) Abstract

5) Figures → should contain the main points

6) Last para of intro → hypothesis & specific investigation

7) Results & discussion → Compare results in that paper to other papers

8) Read Review articles (if new paper)

9) Materials & methods → did they select appropiate methods

10) Supplementary materials

Issues:

+ Logical inconsistency
+ Overgeneralization
+ Question any assumptions
+ Statistical methods

Lecture 7

Monday, 6 February 2023 8:30 AM

Bias and Fallacies : + Confirmation
+ Concomitant variables
+ Bias - randomization
+ Positive & negative controls
+ Factorial design

No magic bands → should be replicable
→ Another student
→ Independent study

Engineering : Design → Build → Test (Not the same as random trial & error)

Learning

Time-scale of research : 12 month plan in quals

Assignment: + No review, opinion, perspective, handbook
+ Primary article/lit → peer reviewed. (Not archive)
+ Take time to explain concepts & abbreviations

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